

A Thermomechanical Theory for Reacting Immiscible Mixtures

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Introduction

An immiscible mixture is one in which the individual constituents of the mixture remain physically separate on some scale. This local separation of the constituents can be incorporated into the construction of a theory for such mixtures in two ways.

First, the kinematic content of the theory can be formulated to model the local structure, or microstructure, of the constituents. In the present work the simplest possible representation of this microstructure is used: in addition to its partial density $\rho_{(\xi)}$, each constituent of the mixture is characterized by its actual, or local, density $\bar{\rho}_{(\xi)} = \rho_{(\xi)}/\varphi_{(\xi)}$, where $\varphi_{(\xi)}$ is the volume of the ξ^{th} constituent per unit volume of the mixture. Furthermore, the volume fractions are subject to the *volume fraction constraint*: the sum of the volume fractions at a spatial point must equal unity.

Second, the constitutive postulates of the theory can reflect the fact that the individual constituents remain separated. Thus, the very general set of constitutive postulates used in this work are tempered by an *immiscibility postulate*: the free energy of a given constituent is assumed to depend only upon the primitive constitutive variables associated with that constituent.*

In a previous work the authors presented a mechanical theory of this type for mixtures of immiscible fluid constituents [9]. The equations of motion of the theory were obtained by the use of Hamilton's extended variational principle. The volume fraction constraint was included in the variational principle and was thereby incorporated directly into the equations of motion. Furthermore, the kinetic energies associated with the local expansion and contraction rates of the constituents were included. The latter terms are important, for example, in the analysis of the dynamic response of bubbly liquids.

The variational principle then yielded the translational equations of motion of the constituents and also the equations governing the local expansion and contraction of the constituents.

In this paper the theory is extended to include thermal effects, solid constituents (such as a porous medium), chemical reactions, and virtual mass effects. The theory is thus raised to the same level of generality that has been achieved for miscible mixtures (see, for example, BOWEN [10]).

Hamilton's principle is first used to derive the equations of motion of the mixture under the assumption that mass transfer between constituents can occur. The equation of motion for the mixture as a whole is derived in addition to the equations for the individual constituents. The equation of balance of energy for each constituent is then derived from the variational principle and the derivation of the balance of energy equation for the mixture as a whole is presented.

A set of constitutive postulates is introduced in which the Helmholtz free energies of the constituents are subject to the immiscibility postulate. The entropy inequality is then used to derive restrictions upon the constitutive postulates.

Finally, the theory is used to obtain models for several types of mixtures that are important in applications.

Among previous works which were the antecedents of this theory, special mention must be made of the work of BIOT [11-15] on fluid-saturated porous media. He derived the equations of motion from a variational principle, and, in [15], included thermal effects. He did not include the volume fractions in his description of the kinematics of the materials and so did not use the volume fraction constraint.

The equations of static equilibrium for a mixture were derived through the use of a variational principle by KENYON [16], ARON [17] and AIZICOVICI &

* From these considerations it is clear that continuum theories for miscible mixtures [1-8] are generally not adequate to treat the types of problems which are addressed in this work. Since those works usually characterize a constituent only by its partial density $\rho_{(s)}$, the actual density $\bar{\rho}_{(s)}$ cannot be recovered. Thus, essential information concerning the local state of the constituent is not available. Furthermore, the volume fraction constraint cannot be used. (See the discussion by BEDFORD & DRUMHELLER [9].)

ARON [18] derived variational principles for miscible mixtures by starting with postulated equations of motion.

The volume fractions of the constituents and the volume fraction constraint were included in the theories of mixtures of immiscible incompressible fluids developed by MILLS [19] and CRAINE [20]. The incompressible solid-fluid case was treated by KENYON [21]. A thermomechanical theory of mixtures of compressible constituents which incorporates volume fractions was presented by PASSMAN [22]; however, he did not use the volume fraction constraint.

A theory of a single continuum which includes the volume fraction as an independent variable was developed by GOODMAN & COWIN [23]. They included the kinetic energy of local expansion of the material; furthermore, their equations of motion were motivated by a variational treatment of the static case [24].

Theories of immiscible mixtures which include very general descriptions of the local structure of the constituents were presented by TWISS & ERINGEN [25, 26], DREW [27], and DREW & SEGEL [28]. Their equations of motion and of balance of energy were obtained by averaging processes applied to the local equations of balance for the individual constituents, rather than by using a variational principle.

1. Preliminaries

Notations and necessary results from the continuum theory of mixtures [10] and the variational methods of mechanics [9, 29–31] are reviewed in this section.

The N constituents of the mixture are modeled by superimposed continua. The motion of the ξ^{th} constituent is given by

$$\mathbf{x} = \mathbf{x}_{(\xi)}(\mathbf{X}_{(\xi)}, t), \quad (1.1)$$

where $\mathbf{x}$ is the spatial position vector of the material particle of the ξ^{th} constituent whose position is $\mathbf{X}_{(\xi)}$ in a specified reference configuration.

The inversion of the motion (1.1) is

$$\mathbf{X}_{(\xi)} = \mathbf{x}_{(\xi)}^{-1}(\mathbf{x}, t). \quad (1.2)$$

The velocity of the ξ^{th} constituent is

$$\mathbf{v}_{(\xi)} = \frac{\partial}{\partial t} \mathbf{x}_{(\xi)}(\mathbf{X}_{(\xi)}, t) \quad (1.3)$$

$$= \hat{\mathbf{v}}_{(\xi)}(\mathbf{X}_{(\xi)}, t). \quad (1.4)$$

The caret will be used to denote functions of $\mathbf{X}_{(\xi)}$ and t (the material representation). The acceleration of the ξ^{th} constituent is

$$\mathbf{a}_{(\xi)} = \frac{\partial}{\partial t} \hat{\mathbf{v}}_{(\xi)}(\mathbf{X}_{(\xi)}, t) = \hat{\mathbf{a}}_{(\xi)}(\mathbf{X}_{(\xi)}, t). \quad (1.5)$$

The material derivative of a function $\Gamma_{(\xi)}(\mathbf{x}, t)$ following the motion of the ξ^{th} constituent is

$$\dot{\Gamma}_{(\xi)}|_{\mathbf{x}_{(\xi)}} = \frac{\partial}{\partial t} \Gamma_{(\xi)} + \text{grad } \Gamma_{(\xi)} \cdot \mathbf{v}_{(\xi)}, \quad (1.6)$$

where grad denotes the spatial gradient. When $\xi = \zeta$, the notation $\dot{\Gamma}_{(\xi)}|_{\mathbf{x}_{(\xi)}}$ will be abbreviated as $\dot{\Gamma}_{(\zeta)}$.

The partial density of the ξ^{th} constituent is denoted by $\rho_{(\xi)}$. In a mixture of immiscible constituents, each constituent has an actual or local density $\bar{\rho}_{(\xi)}$. The volume fraction of the ξ^{th} constituent is then given by

$$\varphi_{(\xi)} = \rho_{(\xi)} / \bar{\rho}_{(\xi)}. \quad (1.7)$$

Note that for compressible constituents ($\bar{\rho}_{(\xi)} \neq \text{const.}$), $\varphi_{(\xi)}$ and $\bar{\rho}_{(\xi)}$ separately can vary independently of $\rho_{(\xi)}$. The volume fractions are assumed to satisfy

$$\sum_{\xi} \varphi_{(\xi)} = 1, \quad (1.8)$$

where $\sum_{\xi}$ denotes the sum from $\xi = 1$ to $\xi = N$.

The Jacobian of the motion (1.1) is

$$J_{(\xi)} = \det \left[\frac{\partial \mathbf{x}_{(\xi)}}{\partial \mathbf{X}_{(\xi)}} \right]. \quad (1.9)$$

where det denotes the determinant.

For the variational development in this work it is convenient to write the equation of conservation of mass of the ξ^{th} constituent in the form

$$\rho_{(\xi)} J_{(\xi)} = \rho_{(\xi)0} + c_{(\xi)}, \quad (1.10)$$

where $\rho_{(\xi)0}$ denotes the partial density of the ξ^{th} constituent in the reference configuration and $c_{(\xi)}$ is the effective increase or decrease of $\rho_{(\xi)0}$ due to mass transfer from the other constituents to the ξ^{th} constituent. Equation (1.10) can be written

$$\dot{\rho}_{(\xi)} + \rho_{(\xi)} \text{div } \mathbf{v}_{(\xi)} = \dot{c}_{(\xi)} / J_{(\xi)}, \quad (1.11)$$

where div denotes the spatial divergence.*

By summing the N equations (1.11), one obtains the equation of conservation of mass for the mixture as a whole,

$$\dot{\rho} + \rho \text{div } \mathbf{v} = \sum_{\xi} \dot{c}_{(\xi)} / J_{(\xi)}, \quad (1.12)$$

where the mixture density ρ is

$$\rho = \sum_{\xi} \rho_{(\xi)}, \quad (1.13)$$

* From (1.11) it can be seen that the term $c_{(\xi)}$ introduced here is related to the mass supply term used in previous works ($\hat{c}$ in BOWEN's notation [10]) by $\dot{c}_{(\xi)} / J_{(\xi)} = \hat{c}_{\xi}$.

and the mixture or barycentric velocity $\mathbf{v}$ is defined by

$$\mathbf{v} = \frac{1}{\rho} \sum_{\xi} \rho_{(\xi)} \mathbf{v}_{(\xi)}. \quad (1.14)$$

In equation (1.12) $\dot{\rho}$ denotes the material derivative of the mixture density following the motion of the mixture. From equation (1.6),

$$\dot{\rho} = \frac{\partial}{\partial t} \rho + \text{grad } \rho \cdot \mathbf{v}. \quad (1.15)$$

Conservation of mass requires that

$$\sum_{\xi} \dot{c}_{(\xi)} / J_{(\xi)} = 0; \quad (1.16)$$

thus (1.12) becomes

$$\dot{\rho} + \rho \text{ div } \mathbf{v} = 0. \quad (1.17)$$

A convenient result derived directly from equation (1.6) is

$$\dot{\Gamma}_{(\xi)} - \dot{\Gamma}_{(\xi)}|_{\mathbf{x}} = \text{grad } \Gamma_{(\xi)} \cdot \mathbf{u}_{(\xi)}, \quad (1.18)$$

where $\dot{\Gamma}_{(\xi)}|_{\mathbf{x}}$ denotes the material derivative following the barycentric motion and where the diffusion velocity $\mathbf{u}_{(\xi)}$ is

$$\mathbf{u}_{(\xi)} = \mathbf{v}_{(\xi)} - \mathbf{v}. \quad (1.19)$$

Furthermore, for the functional Γ defined by

$$\rho \Gamma = \sum_{\xi} \rho_{(\xi)} \Gamma_{(\xi)}, \quad (1.20)$$

equations (1.18) and (1.11) give

$$\rho \dot{\Gamma} = \sum_{\xi} \left[\rho_{(\xi)} \dot{\Gamma}_{(\xi)} - \text{div} (\rho_{(\xi)} \Gamma_{(\xi)} \mathbf{u}_{(\xi)}) + \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} \Gamma_{(\xi)} \right]. \quad (1.21)$$

When a variation $\delta \mathbf{x}_{(\xi)}$ is added to the motion (1.1),

$$\mathbf{x} = \chi_{(\xi)}(\mathbf{X}_{(\xi)}, t) + \delta \mathbf{x}_{(\xi)}(\mathbf{X}_{(\xi)}, t), \quad (1.22)$$

the resulting variations of the velocity and of the Jacobian of the ξ^{th} constituent are

$$\delta \mathbf{v}_{(\xi)} = \delta \dot{\mathbf{x}}_{(\xi)}, \quad (1.23)$$

$$\delta J_{(\xi)} = J_{(\xi)} \text{ div } \delta \mathbf{x}_{(\xi)}. \quad (1.24)$$

The variation of a function $\Gamma_{(\xi)}$ holding the spatial point $\mathbf{x}$ fixed is

$$\delta \Gamma_{(\xi)}|_{\mathbf{x}} = \delta \Gamma_{(\xi)} - \text{grad } \Gamma_{(\xi)} \cdot \delta \mathbf{x}_{(\xi)}, \quad (1.25)$$

where $\delta I_{(\xi)}$ denotes the usual variation holding the material particle of the ξ^{th} constituent fixed. Similarly, the variation of $I_{(\xi)}$ holding the material particle of the ζ^{th} constituent fixed is

$$\delta I_{(\xi)}|_{\mathbf{x}_{(\xi)}} = \delta I_{(\xi)} - \text{grad } I_{(\xi)} \cdot (\delta \mathbf{x}_{(\xi)} - \delta \mathbf{x}_{(\zeta)}). \quad (1.26)$$

For a material volume v of the ξ^{th} constituent,

$$\delta \int_v I_{(\xi)} dv = \int_v \delta I_{(\xi)}|_{\mathbf{x}} dv, \quad (1.27)$$

and it is easy to show that

$$\delta \int_v \rho_{(\xi)} I_{(\xi)} dv = \int_v (\rho_{(\xi)} \delta I_{(\xi)} + I_{(\xi)} \delta c_{(\xi)}/J_{(\xi)}) dv. \quad (1.28)$$

Finally, Hamilton's extended principle requires that the motion of the mixture satisfy

$$\int_{t_1}^{t_2} (\delta T - \delta U + \delta W) dt = 0 \quad (1.29)$$

for arbitrary times t_1 and t_2 , where T and U are the total kinetic and potential energies of the mixture and where δW is the virtual work of forces not represented by U .

2. The Variational Principle and the Equations of Motion

The equations of motion are obtained by applying Hamilton's extended principle (1.29) to a material volume v of the mixture whose surface s is a specified function of time [9]. It is assumed that the variations $\delta \mathbf{x}_{(\xi)}$ vanish at s .

The total kinetic energy of the mixture in v is written

$$\begin{aligned} T = & \sum_{\xi} \int_v \frac{1}{2} \rho_{(\xi)} \mathbf{v}_{(\xi)} \cdot \mathbf{v}_{(\xi)} dv \\ & + \sum_{\xi} \sum_{\zeta} \int_v \frac{1}{2} m_{(\xi\zeta)} (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) \cdot (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) dv \\ & + \sum_{\xi} \int_v \frac{1}{2} \rho_{(\xi)} I_{(\xi)} \dot{\bar{\rho}}_{(\xi)}^2 dv. \end{aligned} \quad (2.1)$$

The first integral in (2.1) is the total translational kinetic energy of the constituents. The second integral introduces the virtual mass terms (see Appendix A). The third integral is the total kinetic energy of local expansion and contraction of the constituents [9]. The coefficients $I_{(\xi)}$ and $m_{(\xi\zeta)}$ are assumed to be functions of $\bar{\rho}_{(\xi)}$, $\varphi_{(\xi)}$, $\zeta = 1, 2, \dots, N$.

In the present formulation all mechanical forces will be introduced as generalized forces in the expression for virtual work:

$$\begin{aligned}
 \delta W = & \sum_{\xi} \int_v -\text{tr}(\mathbf{T}_{(\xi)}^t \text{grad } \delta \mathbf{x}_{(\xi)}) dv \\
 & + \sum_{\xi} \int_v -\varphi_{(\xi)} P_{(\xi)} \frac{\delta \bar{\rho}_{(\xi)}}{\bar{\rho}_{(\xi)}} dv \\
 & + \sum_{\xi} \int_v L_{(\xi)} \frac{\delta c_{(\xi)}}{J_{(\xi)}} dv \\
 & + \sum_{\xi} \int_v (\rho_{(\xi)} \mathbf{f}_{(\xi)} + \mathbf{b}_{(\xi)}) \cdot \delta \mathbf{x}_{(\xi)} dv,
 \end{aligned} \tag{2.2}$$

where tr denotes the trace and superscript t denotes the transpose. The first integral in (2.2) is the virtual work due to the virtual deformation and rotation of the ξ^{th} constituent. The associated generalized force $\mathbf{T}_{(\xi)}$ is a second-order tensor. The remaining three integrals are the virtual works due to the variations $\delta \bar{\rho}_{(\xi)}$, $\delta c_{(\xi)}$, and $\delta \mathbf{x}_{(\xi)}$. The generalized force associated with the variation $\delta \mathbf{x}_{(\xi)}$ is split into the external body force density $\mathbf{f}_{(\xi)}$ and the momentum supply $\mathbf{b}_{(\xi)}$ due to interactions between constituents. More precise interpretations of the generalized forces $\mathbf{T}_{(\xi)}$, $P_{(\xi)}$, $L_{(\xi)}$, and $\mathbf{b}_{(\xi)}$ will become possible as the development proceeds.

Before the application of Hamilton's principle, several constraints on the variations will be introduced through the method of Lagrange multipliers. The constraint equations are (1.8), (1.10), and (1.16), to which are assigned Lagrange multipliers λ , $\mu_{(\xi)}$, and τ , respectively.

Hamilton's principle for the mixture is therefore

$$\begin{aligned}
 \delta \int_{t_1}^{t_2} \left\{ \sum_{\xi} \int_v \frac{1}{2} \rho_{(\xi)} \mathbf{v}_{(\xi)} \cdot \mathbf{v}_{(\xi)} dv \right. \\
 + \sum_{\xi} \sum_{\zeta} \int_v \frac{1}{2} m_{(\xi\zeta)} (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) \cdot (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) dv \\
 + \sum_{\xi} \int_v \frac{1}{2} \rho_{(\xi)} I_{(\xi)} \dot{\bar{\rho}}_{(\xi)}^2 dv \Big\} dt \\
 + \int_{t_1}^{t_2} \left\{ \sum_{\xi} \int_v \left[-\text{tr}(\mathbf{T}_{(\xi)}^t \text{grad } \delta \mathbf{x}_{(\xi)}) - \varphi_{(\xi)} P_{(\xi)} \frac{\delta \bar{\rho}_{(\xi)}}{\bar{\rho}_{(\xi)}} \right. \right. \\
 + L_{(\xi)} \frac{\delta c_{(\xi)}}{J_{(\xi)}} + (\rho_{(\xi)} \mathbf{f}_{(\xi)} + \mathbf{b}_{(\xi)}) \cdot \delta \mathbf{x}_{(\xi)} \Big] dv \\
 - \int_v \lambda \delta \left(\sum_{\xi} \varphi_{(\xi)} - 1 \right) \Big|_x dv + \int_v \tau \delta \left(\sum_{\xi} \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} \right) \Big|_x dv \\
 \left. + \sum_{\xi} \int_v \mu_{(\xi)} \delta \left(J_{(\xi)} - \frac{\rho_{(\xi)0} + c_{(\xi)}}{\rho_{(\xi)}} \right) dv \right\} dt = 0.
 \end{aligned} \tag{2.4}$$

The variations of several of the terms in (2.4) are derived in Appendices A and B. When the results of those manipulations are substituted into (2.4), the coefficients of $\delta \mathbf{x}_{(\xi)}$, $\delta \bar{\rho}_{(\xi)}$, $\delta \varphi_{(\xi)}$, and $\delta c_{(\xi)}$ can be equated to zero to obtain the constituent equations of motion:

$$\begin{aligned}
\rho_{(\xi)} \mathbf{a}_{(\xi)} = & -\frac{\dot{c}_{(\xi)}}{J_{(\xi)}} \mathbf{v}_{(\xi)} + \operatorname{div} \mathbf{T}_{(\xi)} + \rho_{(\xi)} \mathbf{f}_{(\xi)} \\
& + \mathbf{b}_{(\xi)} + \lambda \operatorname{grad} \varphi_{(\xi)} - \operatorname{grad} (\mu_{(\xi)} J_{(\xi)}) \\
& + \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} \operatorname{grad} \tau - \sum_{\zeta} (m_{(\xi\zeta)} + m_{(\zeta\xi)}) [\mathbf{a}_{(\xi)} - \mathbf{a}_{(\zeta)} \\
& + (\mathcal{L}_{(\xi)}^t - \mathcal{L}_{(\xi)}) (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) + (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) \operatorname{div} \mathbf{v}_{(\xi)}] \\
& - \sum_{\zeta} \overline{(m_{(\xi\zeta)} + m_{(\zeta\xi)})} |_{\mathbf{x}_{(\xi)}} (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) \\
& - \frac{1}{2} \sum_{\zeta} \sum_{\gamma} \left[\frac{\partial m_{(\gamma\zeta)}}{\partial \bar{\rho}_{(\xi)}} \operatorname{grad} \bar{\rho}_{(\xi)} + \frac{\partial m_{(\gamma\zeta)}}{\partial \varphi_{(\xi)}} \operatorname{grad} \varphi_{(\xi)} \right] (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\zeta)}) \cdot (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\zeta)}) \\
& + \frac{1}{2} \rho_{(\xi)} (\operatorname{grad} I_{(\xi)}) \dot{\bar{\rho}}_{(\xi)}^2 \\
& - \sum_{\zeta} \frac{1}{2} \rho_{(\zeta)} \frac{\partial I_{(\zeta)}}{\partial \bar{\rho}_{(\xi)}} \dot{\bar{\rho}}_{(\xi)}^2 \operatorname{grad} \bar{\rho}_{(\xi)} \\
& - \sum_{\zeta} \frac{1}{2} \rho_{(\zeta)} \frac{\partial I_{(\zeta)}}{\partial \varphi_{(\xi)}} \dot{\bar{\rho}}_{(\xi)}^2 \operatorname{grad} \varphi_{(\xi)}, \tag{2.5}
\end{aligned}$$

$$\begin{aligned}
\bar{\rho}_{(\xi)}^2 \overline{(I_{(\xi)} \dot{\bar{\rho}}_{(\xi)})} = & -\frac{\bar{\rho}_{(\xi)}}{\varphi_{(\xi)}} \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} I_{(\xi)} \dot{\bar{\rho}}_{(\xi)} + \frac{\mu_{(\xi)} J_{(\xi)}}{\varphi_{(\xi)}} - P_{(\xi)} \\
& + \frac{1}{2} \frac{\bar{\rho}_{(\xi)}}{\varphi_{(\xi)}} \sum_{\zeta} \sum_{\gamma} \frac{\partial m_{(\gamma\zeta)}}{\partial \rho_{(\xi)}} (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\zeta)}) \cdot (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\zeta)}) \\
& + \frac{1}{2} \frac{\bar{\rho}_{(\xi)}}{\varphi_{(\xi)}} \sum_{\zeta} \rho_{(\zeta)} \frac{\partial I_{(\zeta)}}{\partial \rho_{(\xi)}} \bar{\rho}_{(\xi)}^2, \tag{2.6}
\end{aligned}$$

$$\begin{aligned}
& -\lambda + \frac{\mu_{(\xi)} J_{(\xi)}}{\varphi_{(\xi)}} + \frac{1}{2} \sum_{\zeta} \rho_{(\zeta)} \frac{\partial I_{(\zeta)}}{\partial \varphi_{(\xi)}} \dot{\bar{\rho}}_{(\xi)}^2 \\
& + \frac{1}{2} \sum_{\zeta} \sum_{\gamma} \frac{\partial m_{(\gamma\zeta)}}{\partial \varphi_{(\xi)}} (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\zeta)}) \cdot (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\zeta)}) = 0, \tag{2.7}
\end{aligned}$$

$$\frac{1}{2} \mathbf{v}_{(\xi)} \cdot \mathbf{v}_{(\xi)} + \frac{1}{2} I_{(\xi)} \dot{\bar{\rho}}_{(\xi)}^2 + L_{(\xi)} - \frac{\mu_{(\xi)} J_{(\xi)}}{\rho_{(\xi)}} = \dot{\tau} |_{\mathbf{x}_{(\xi)}}, \tag{2.8}$$

where $\mathcal{L}_{(\xi)} = \frac{\partial \mathbf{v}_{(\xi)}}{\partial \mathbf{x}}$.

In closing this section the equation of balance of linear momentum for the mixture as a whole will be derived from (2.5). By noting (1.14) and (1.20) and substituting $\mathbf{f}_{(\xi)} = \mathbf{v}_{(\xi)}$ into (1.21), one obtains

$$\rho \mathbf{a} = \sum_{\xi} \left[\rho_{(\xi)} \mathbf{a}_{(\xi)} - \operatorname{div} (\rho_{(\xi)} \mathbf{u}_{(\xi)} \otimes \mathbf{u}_{(\xi)}) + \frac{c_{(\xi)}}{J_{(\xi)}} \mathbf{u}_{(\xi)} \right]. \tag{2.9}$$

Then by making the assumption that

$$\sum_{\xi} \mathbf{b}_{(\xi)} = \mathbf{0} \tag{2.10}$$

manipulations

$$\rho E = \text{tr}(\mathbf{T}' \mathcal{L}) - \text{div} \mathbf{q} + \rho r, \quad (3.4)$$

where

$$\begin{aligned} \rho E = & \sum_{\xi} \rho_{(\xi)} (E_{(\xi)} + \frac{1}{2} \mathbf{u}_{(\xi)} \cdot \mathbf{u}_{(\xi)} + \frac{1}{2} I_{(\xi)} \bar{\rho}_{(\xi)}^2) \\ & + \sum_{\xi} \sum_{\zeta} \frac{1}{4} (m_{(\xi\zeta)} + m_{(\zeta\xi)}) (\mathbf{u}_{(\xi)} - \mathbf{u}_{(\zeta)}) \cdot (\mathbf{u}_{(\xi)} - \mathbf{u}_{(\zeta)}), \end{aligned} \quad (3.5)$$

$$\begin{aligned} \mathbf{q} = & \sum_{\xi} \{ \mathbf{q}_{(\xi)} - (\mathbf{T}'_{(\xi)} - \mu_{(\xi)} J_{(\xi)} \mathbf{l}) \mathbf{u}_{(\xi)} \\ & + \rho_{(\xi)} (E_{(\xi)} + \frac{1}{2} I_{(\xi)} \bar{\rho}_{(\xi)}^2 + \frac{1}{2} \mathbf{u}_{(\xi)} \cdot \mathbf{u}_{(\xi)}) \mathbf{u}_{(\xi)} \\ & + \sum_{\zeta} \frac{1}{2} (m_{(\xi\zeta)} + m_{(\zeta\xi)}) [(\mathbf{u}_{(\xi)} - \mathbf{u}_{(\zeta)}) \otimes (\mathbf{u}_{(\xi)} - \mathbf{u}_{(\zeta)}) \mathbf{u}_{(\xi)} \\ & + (\mathbf{u}_{(\xi)} - \mathbf{u}_{(\zeta)}) \cdot (\mathbf{u}_{(\xi)} - \mathbf{u}_{(\zeta)}) \mathbf{u}_{(\xi)}] \}, \end{aligned} \quad (3.6)$$

and

$$\rho r = \sum_{\xi} (\rho_{(\xi)} r_{(\xi)} + \rho_{(\xi)} \mathbf{f}_{(\xi)} \cdot \mathbf{u}_{(\xi)}). \quad (3.7)$$

Equations (3.5), (3.6) and (3.7) define the mixture internal energy, heat flux, and specific external heat supply.

Again, it is seen that the mixture energy relation (3.4) is identical to the conventional relation for a single continuum. Also note that the virtual mass terms affect the definitions of $\mathbf{T}$, E , and $\mathbf{q}$.

4. Constitutive Postulates and the Thermodynamic Restrictions

To obtain the complete theory the balance of mass equations (1.11) and (1.16), the balance of momentum equations (2.5)–(2.8) and (2.10), and the balance of energy equations (3.2) and (3.3) must be supplemented with appropriate constitutive postulates.* The first step in stating the constitutive postulates is to choose a set of primitive or independent variables for the constitutive functions.

In this work the local densities $\bar{\rho}_{(\xi)}$, the deformation gradients $\mathbf{F}_{(\xi)} = \frac{\partial \mathbf{x}_{(\xi)}}{\partial \mathbf{X}_{(\xi)}}$, the temperatures $\theta_{(\xi)}$, the temperature gradients $\mathbf{g}_{(\xi)} = \text{grad } \theta_{(\xi)}$, and the density gradients $\mathbf{G}_{(\xi)} = \text{grad } \bar{\rho}_{(\xi)}$ together with the rates $\dot{\bar{\rho}}_{(\xi)}$, $\dot{\mathbf{F}}_{(\xi)}$, $\mathbf{v}_{(\xi)}$, and $\dot{c}_{(\xi)}$ will be used.**

* This section closely follows the procedures that are used in the theory of miscible mixtures [10].

** It has been traditional in works on chemically reacting mixtures to write constitutive equations for $\dot{c}_{(\xi)}$ [35]. In the present formulation, it seems most consistent to include $\dot{c}_{(\xi)}$ among the primitive variables and to write constitutive equations for $L_{(\xi)}$, the generalized forces that are conjugate to the $\delta c_{(\xi)}$ in the virtual work expression (2.2). In this way the rate of work expression (3.1) is seen to consist of products of generalized force terms which are constitutive and rate terms which are primitive. Further indications that this is the most consistent choice will be seen when the thermodynamic restrictions upon the constitutive functions are derived.

The constitutive postulates can then be stated. Since the constituent temperatures $\theta_{(\xi)}$ are chosen as primitive variables of the mixture, it is convenient to use the Helmholtz free energy

$$\Psi_{(\xi)} = E_{(\xi)} - \theta_{(\xi)} \eta_{(\xi)}, \quad (4.1)$$

where $\eta_{(\xi)}$ is the specific entropy of the ξ^{th} constituent, in place of the specific internal energy $E_{(\xi)}$.

In writing the constitutive functions for $\Psi_{(\xi)}$, we have used the immiscibility postulate. We assume that the free energy of the ξ^{th} constituent is completely determined by the primitive constitutive variables of that constituent alone. This assumption provides a formal definition of the term *immiscible mixture*. Therefore the Helmholtz free energy is written

$$\Psi_{(\xi)} = \Psi_{(\xi)}(\bar{\rho}_{(\xi)}, \dot{\bar{\rho}}_{(\xi)}, \mathbf{G}_{(\xi)}, \mathbf{F}_{(\xi)}, \dot{\mathbf{F}}_{(\xi)}, \mathbf{v}_{(\xi)}, \dot{c}_{(\xi)}, \theta_{(\xi)}, \mathbf{g}_{(\xi)}), \quad (4.2)$$

where $\xi = 1, 2, \dots, N$.

The remaining dependent constitutive variables are assumed to be functions of the primitive variables of all of the constituents. In abbreviated form,

$$\begin{aligned} & (P_{(\xi)}, \mathbf{T}_{(\xi)}, \mathbf{q}_{(\xi)}, L_{(\xi)}, \mathbf{b}_{(\xi)}, \dot{e}_{(\xi)}, \eta_{(\xi)}) \\ & = f(\bar{\rho}_{(\xi)}, \dot{\bar{\rho}}_{(\xi)}, \mathbf{G}_{(\xi)}, \dot{\mathbf{F}}_{(\xi)}, \mathbf{F}_{(\xi)}, \mathbf{v}_{(\xi)}, \dot{c}_{(\xi)}, \theta_{(\xi)}, \mathbf{g}_{(\xi)}), \end{aligned} \quad (4.3)$$

where $\xi, \zeta = 1, 2, \dots, N$. Note that only $N - 1$ constitutive relations are written for $\mathbf{b}_{(\xi)}$ and for $\dot{e}_{(\xi)}$ due to (2.10) and (3.3).

Once the constitutive functions are postulated, equations (1.8), (1.16), (2.10) and (3.3), the equations of balance of mass (1.11), the equations of motion (2.5), (2.6), (2.7), and (2.8), the equations of balance of energy (3.2), and the constitutive equations (4.2) and (4.3) are a system of $14N + 2$ equations in the $14N + 2$ unknowns $\bar{\rho}_{(\xi)}, \varphi_{(\xi)}, \mathbf{v}_{(\xi)}, c_{(\xi)}, P_{(\xi)}, \mathbf{T}_{(\xi)}, L_{(\xi)}, \mathbf{b}_{(\xi)}, \lambda, \tau, \mu_{(\xi)}, \mathbf{q}_{(\xi)}, \dot{e}_{(\xi)}, \Psi_{(\xi)}, \theta_{(\xi)}$, and $\eta_{(\xi)}$.

The relations (4.2) and (4.3) are subject to the restrictions imposed by equations (1.16), (2.10) and (3.3) and also to the restrictions derived from the entropy inequality by the method of COLEMAN & NOLL [36]. The BOWEN-MÜLLER entropy inequality for a mixture is [10]

$$\sum_{\xi} \left[\rho_{(\xi)} \dot{\eta}_{(\xi)} + \text{div}(\mathbf{q}_{(\xi)}/\theta_{(\xi)}) - \rho_{(\xi)} r_{(\xi)}/\theta_{(\xi)} + \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} \eta_{(\xi)} \right] \geq 0. \quad (4.4)$$

By solving (3.2) for $r_{(\xi)}$ and substituting it together with (4.1) into (4.4), and then expanding $\dot{\Psi}_{(\xi)}$ in terms of the primitive variables by using the chain rule, one obtains

$$\begin{aligned} & \sum_{\xi} \left\{ -\frac{\rho_{(\xi)}}{\theta_{(\xi)}} \left(\eta_{(\xi)} + \frac{\partial \Psi_{(\xi)}}{\partial \theta_{(\xi)}} \right) \theta_{(\xi)} - \frac{\rho_{(\xi)}}{\theta_{(\xi)}} \frac{\partial \Psi_{(\xi)}}{\partial \dot{\bar{\rho}}_{(\xi)}} \dot{\bar{\rho}}_{(\xi)} \right. \\ & \quad \left. - \frac{\rho_{(\xi)}}{\theta_{(\xi)}} \frac{\partial \Psi_{(\xi)}}{\partial \mathbf{v}_{(\xi)}} \cdot \mathbf{a}_{(\xi)} \right\} \end{aligned}$$

$$\begin{aligned}
& -\frac{\rho_{(\xi)}}{\theta_{(\xi)}} \operatorname{tr} \left(\frac{\partial \Psi_{(\xi)}}{\partial \dot{\mathbf{F}}_{(\xi)}} \dot{\mathbf{F}}_{(\xi)} \right) - \frac{\rho_{(\xi)}}{\theta_{(\xi)}} \frac{\partial \Psi_{(\xi)}}{\partial \mathbf{g}_{(\xi)}} \cdot \dot{\mathbf{g}}_{(\xi)} \\
& - \frac{\rho_{(\xi)}}{\theta_{(\xi)}} \frac{\partial \Psi_{(\xi)}}{\partial \dot{\mathbf{G}}_{(\xi)}} \cdot \dot{\mathbf{G}}_{(\xi)} - \frac{\rho_{(\xi)}}{\theta_{(\xi)}} \frac{\partial \Psi_{(\xi)}}{\partial \dot{c}_{(\xi)}} \dot{c}_{(\xi)} \\
& + \frac{\varphi_{(\xi)}}{\theta_{(\xi)}} \left(P_{(\xi)} - \bar{\rho}_{(\xi)}^2 \frac{\partial \Psi_{(\xi)}}{\partial \bar{\rho}_{(\xi)}} \right) \frac{\dot{\bar{\rho}}_{(\xi)}}{\bar{\rho}_{(\xi)}} \\
& + \operatorname{tr} \left[\left(\mathbf{T}_{(\xi)}^t - \rho_{(\xi)} \mathbf{F}_{(\xi)} \left(\frac{\partial \Psi_{(\xi)}}{\partial \mathbf{F}_{(\xi)}} \right)' \right) \frac{\mathcal{L}_{(\xi)}}{\theta_{(\xi)}} \right] - \frac{\mathbf{q}_{(\xi)} \cdot \mathbf{g}_{(\xi)}}{\theta_{(\xi)}^2} \\
& - \frac{1}{\theta_{(\xi)}} \left(\Psi_{(\xi)} + L_{(\xi)} \right) \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} + \frac{\dot{\theta}_{(\xi)}}{\theta_{(\xi)}} \Big\} \geq 0.
\end{aligned} \tag{4.5}$$

The coefficients of those terms in (4.5) which can be varied independently must vanish in order that the inequality be satisfied for any admissible process. As shown in Appendix C, after accounting for the constraint conditions (1.8), (1.10) and (1.16) and their gradients and time derivatives, the inequality (4.5) requires that $\Psi_{(\xi)}$ cannot depend upon $\bar{\rho}_{(\xi)}$, $\mathbf{v}_{(\xi)}$, $\dot{\mathbf{F}}_{(\xi)}$, $\mathbf{g}_{(\xi)}$, $\dot{\mathbf{G}}_{(\xi)}$, and $\dot{c}_{(\xi)}$; that is,

$$\Psi_{(\xi)} = \Psi_{(\xi)}(\bar{\rho}_{(\xi)}, \mathbf{F}_{(\xi)}, \theta_{(\xi)}), \tag{4.6}$$

and furthermore,

$$\eta_{(\xi)} = -\frac{\partial \Psi_{(\xi)}}{\partial \theta_{(\xi)}}. \tag{4.7}$$

Next, in order to develop a set of conditions which are valid in equilibrium, it is convenient to introduce the constraint equations (1.16), (2.10), and (3.3) explicitly into the inequality (4.5) to obtain

$$\begin{aligned}
& \sum_{\xi} \frac{\varphi_{(\xi)}}{\theta_{(\xi)}} \left(P_{(\xi)} - \bar{\rho}_{(\xi)}^2 \frac{\partial \Psi_{(\xi)}}{\partial \bar{\rho}_{(\xi)}} \right) \frac{\dot{\bar{\rho}}_{(\xi)}}{\bar{\rho}_{(\xi)}} \\
& + \sum_{\xi} \operatorname{tr} \left\{ \frac{1}{\theta_{(\xi)}} \left[\mathbf{T}_{(\xi)}^t - \rho_{(\xi)} \mathbf{F}_{(\xi)} \left(\frac{\partial \Psi_{(\xi)}}{\partial \mathbf{F}_{(\xi)}} \right)' \right] \frac{\mathcal{L}_{(\xi)}}{\theta_{(\xi)}} \right\} - \sum_{\xi} \frac{1}{\theta_{(\xi)}^2} \mathbf{q}_{(\xi)} \cdot \mathbf{g}_{(\xi)} \\
& + \sum_{\xi}^{N-1} \left\{ \frac{1}{\theta_{(N)}} (\Psi_{(N)} + L_{(N)}) - \frac{1}{\theta_{(\xi)}} (\Psi_{(\xi)} + L_{(\xi)}) \right\} \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} \\
& + \sum_{\xi}^{N-1} \left\{ \left(\frac{1}{\theta_{(\xi)}} - \frac{1}{\theta_{(N)}} \right) \dot{\theta}_{(\xi)} + \frac{1}{\theta_{(N)}} \mathbf{b}_{(\xi)} \cdot (\mathbf{v}_{(N)} - \mathbf{v}_{(\xi)}) \right\} \geq 0,
\end{aligned} \tag{4.8}$$

where N denotes the (arbitrary) N^{th} constituent.

The inequality (4.8) is equal to zero at the equilibrium state

$$\dot{\bar{\rho}}_{(\xi)} = \mathbf{g}_{(\xi)} = \dot{c}_{(\xi)} = 0, \tag{4.9}$$

$$\mathbf{v}_{(\xi)} = \mathbf{v}_{(N)} = \text{const.}, \tag{4.10}$$

$$\theta_{(\xi)} = \theta_{(N)} = \text{const.}, \tag{4.11}$$

where $\xi = 1, 2, \dots, N$. The left side of (4.8) is also a minimum at this state [34].

Constraints which are imposed upon the equilibrium state by the inequality (4.8) can be obtained by expressing each primitive variable as the sum of its value in the equilibrium state and a perturbation. For example, for the density rates and the velocities,

$$\dot{\bar{\rho}}_{(\xi)} = \beta \dot{\bar{\rho}}_{(\xi)p}, \quad (4.12)$$

$$\mathbf{v}_{(\xi)} = \mathbf{v}_{(N)e} + \beta \mathbf{v}_{(\xi)p}, \quad (4.13)$$

where the subscripts e and p denote the equilibrium and perturbation values respectively, and β is a scalar parameter [35]. When the primitive variables in (4.8) are written as expressions of the form of (4.12) and (4.13), the partial derivative of (4.8) with respect to β must equal zero at the equilibrium point $\beta = 0$, yielding the equation

$$\begin{aligned} & \sum_{\xi} \left\{ \frac{\varphi_{(\xi)e}}{\theta_{(N)e}} \left(P_{(\xi)} - \bar{\rho}_{(\xi)e}^2 \frac{\partial \Psi_{(\xi)}}{\partial \bar{\rho}_{(\xi)}} \right) \frac{\dot{\bar{\rho}}_{(\xi)p}}{\bar{\rho}_{(\xi)e}} \right\} \\ & + \sum_{\xi} \text{tr} \left\{ \frac{1}{\theta_{(N)e}} \left[\mathbf{T}_{(\xi)}^t - \rho_{(\xi)e} \mathbf{F}_{(\xi)e} \left(\frac{\partial \Psi_{(\xi)}}{\partial \mathbf{F}_{(\xi)}} \right)^t \right] \mathcal{L}_{(\xi)} \right\} - \sum_{\xi} \frac{1}{\theta_{(N)e}^2} \mathbf{q}_{(\xi)} \cdot \mathbf{g}_{(\xi)p} \\ & + \sum_{\xi}^{N-1} \frac{1}{\theta_{(N)e}} (\Psi_{(N)} + L_{(N)} - \Psi_{(\xi)} - L_{(\xi)}) \frac{\dot{\xi}_{(\xi)p}}{J_{(\xi)e}} \\ & + \sum_{\xi}^{N-1} \left\{ \frac{1}{\theta_{(N)e}^2} (\theta_{(N)p} - \theta_{(\xi)p}) \dot{\xi}_{(\xi)} + \frac{1}{\theta_{(N)e}} \mathbf{b}_{(\xi)} \cdot (\mathbf{v}_{(N)p} - \mathbf{v}_{(\xi)p}) \right\} = 0. \end{aligned} \quad (4.14)$$

Since the perturbation values of the primitive variables appearing in (4.14) are arbitrary, it can be concluded from (4.14) that, in the equilibrium state,

$$\Psi_{(\xi)} + L_{(\xi)} = \Psi_{(N)} + L_{(N)}, \quad (4.15)$$

$$P_{(\xi)} = \bar{\rho}_{(\xi)e}^2 \frac{\partial \Psi_{(\xi)}}{\partial \bar{\rho}_{(\xi)}}, \quad (4.16)$$

$$\mathbf{T}_{(\xi)} = \rho_{(\xi)e} \frac{\partial \Psi_{(\xi)}}{\partial \mathbf{F}_{(\xi)}} \mathbf{F}_{(\xi)e}^t, \quad (4.17)$$

$$\mathbf{q}_{(\xi)} = 0, \quad (4.18)$$

$$\dot{\xi}_{(\xi)} = 0, \quad (4.19)$$

$$\mathbf{b}_{(\xi)} = 0, \quad (4.20)$$

where $\xi = 1, 2, \dots, N$.

Furthermore, by writing (2.6), (2.7), and (2.8) for the equilibrium state, one obtains

$$\mu_{(\xi)} = \varphi_{(\xi)e} P_{(\xi)} / J_{(\xi)e}, \quad (4.21)$$

$$\lambda = -J_{(\xi)e} \mu_{(\xi)} / \varphi_{(\xi)e}, \quad (4.22)$$

$$L_{(\xi)} = P_{(\xi)} / \bar{\rho}_{(\xi)e}, \quad (4.23)$$

so that the equilibrium values of the generalized forces $P_{(\xi)}$, $T_{(\xi)}$, and $L_{(\xi)}$ and of the multipliers $\mu_{(\xi)}$ and λ are derivable from $\Psi_{(\xi)}$.

Denoting the chemical potential by $g_{(\xi)} = E_{(\xi)} - \theta_{(\xi)} \eta_{(\xi)} + P_{(\xi)} / \bar{\rho}_{(\xi)}$, we note from (4.13) and (4.22) that

$$g_{(\xi)} = g_{(N)}, \quad (4.24)$$

so that the chemical potential of each constituent must have the same value at equilibrium. This is in agreement with classical thermochemistry [37].

For many immiscible mixtures of practical interest, a less general list of primitive variables than has been used in (4.3) would be adequate. In cases where the $\dot{c}_{(\zeta)}$ are not included as primitive variables, it is easy to show that (4.15) applies in any process rather than only in equilibrium. Similarly if the $\dot{\bar{\rho}}_{(\zeta)}$ are not included, then (4.16) applies in any process. If the $\dot{F}_{(\zeta)}$ are omitted, the same conclusion can be made regarding (4.17).

Examples of mixtures which can be modeled by sets of primitive variables less general than (4.3) are described in the following section.

5. Some Special Mixtures

Some specific simplifications of the general theory will be examined to obtain formulations for some common types of mixtures which are important for applications. Five examples will be presented: the sedimentation of rigid particles in an incompressible liquid, a reacting bubbly liquid, a reacting dusty gas, a reacting fluid-saturated porous solid, and a mixture of ideal gases.

Sedimentation of Rigid Particles in an Incompressible Liquid

In this example a distribution of rigid particles settling in an incompressible liquid is modeled by treating the particles and the liquid as fluid continua. Chemical reactions, thermal effects, and the virtual mass terms will be omitted. The primitive variables reduce to $J_{(\xi)}$, $\dot{F}_{(\xi)}$, and $v_{(\xi)}$ where the dependence on the deformation gradient has been limited to the determinant of the deformation gradient. Equations (2.5) and (2.7) become

$$\begin{aligned} \rho_{(\xi)} \mathbf{a}_{(\xi)} &= \rho_{(\xi)} \mathbf{f}_{(\xi)} + \mathbf{b}_{(\xi)} + \lambda \operatorname{grad} \varphi_{(\xi)} \\ &\quad - \operatorname{grad} (\mu_{(\xi)} J_{(\xi)}) + \operatorname{div} \mathbf{T}_{(\xi)} \end{aligned} \quad (5.1)$$

$$-\lambda + \mu_{(\xi)} J_{(\xi)} / \varphi_{(\xi)} = 0. \quad (5.2)$$

By substituting $\mu_{(\xi)} J_{(\xi)}$ from (5.2) into (5.1), one obtains

$$\rho_{(\xi)} \mathbf{a}_{(\xi)} = -\varphi_{(\xi)} \operatorname{grad} \lambda + \operatorname{div} \mathbf{T}_{(\xi)} + \rho_{(\xi)} \mathbf{f}_{(\xi)} + \mathbf{b}_{(\xi)}. \quad (5.3)$$

From (4.3)

$$\mathbf{T}_{(\xi)} = \mathbf{T}_{(\xi)}(J_{(\xi)}, \dot{F}_{(\xi)}, v_{(\xi)}), \quad (5.4)$$

$$\mathbf{b}_{(\xi)} = \mathbf{b}_{(\xi)}(J_{(\xi)}, \dot{F}_{(\xi)}, v_{(\xi)}), \quad (5.5)$$

and from (4.17)

$$\begin{aligned} T_{(\xi)e} &= \rho_{(\xi)e} \frac{\partial \Psi_{(\xi)}}{\partial J_{(\xi)}} \frac{\partial J_{(\xi)}}{\partial \mathbf{F}_{(\xi)}} \mathbf{F}_{(\xi)e}^T \\ &= \rho_{(\xi)e} J_{(\xi)e} \frac{\partial \Psi_{(\xi)}}{\partial J_{(\xi)}} \mathbf{I}. \end{aligned} \quad (5.6)$$

Furthermore, since the $\bar{\rho}_{(\xi)}$ are constant and $c_{(\xi)}=0$, (1.10) requires that the product $\varphi_{(\xi)} J_{(\xi)}$ be a constant. Consequently $\Psi_{(\xi)}$ can be considered a function of $\varphi_{(\xi)}$ instead of $J_{(\xi)}$, which gives

$$T_{(\xi)e} = -\bar{\rho}_{(\xi)e} \varphi_{(\xi)e}^2 \frac{\partial \Psi_{(\xi)}}{\partial \varphi_{(\xi)}} \mathbf{I}. \quad (5.7)$$

Equations (1.8), (1.11), (5.3), (5.4), and (5.5) are a system of equations in the unknowns λ , $\varphi_{(\xi)}$, $\mathbf{v}_{(\xi)}$, $\mathbf{T}_{(\xi)}$, and $\mathbf{h}_{(\xi)}$.

When the $\mathbf{T}_{(\xi)}$ are omitted, (1.8), (1.11), and (5.3)–(5.5) are identical to the equations for particulate sedimentation presented by THACKER & LAVELLE [38]. This case was also treated in [9]. The terms $\mathbf{T}_{(\xi)}$ introduce into the theory the viscosity of the fluid and the diffusive effects due to Brownian motions of the particles.*

If $\dot{\mathbf{F}}_{(\xi)}$ is omitted from the primitive variables, it is interesting to note that the entropy inequality (4.8) reduces to

$$\mathbf{h}_{(s)} \cdot (\mathbf{v}_{(f)} - \mathbf{v}_{(s)}) \geq 0, \quad (5.8)$$

where s denotes the particles and f denotes the liquid.

The Bubbly Liquid

To model an inviscid liquid containing a distribution of bubbles it is assumed that

$$\Psi_{(\xi)} = \Psi_{(\xi)}(\bar{\rho}_{(\xi)}, \theta_{(\xi)}). \quad (5.9)$$

In addition, $\mathbf{G}_{(\xi)}$ and $\dot{\mathbf{F}}_{(\xi)}$ are excluded from the list of primitive variables, so that from (4.16) and (4.17)

$$P_{(\xi)e} = \bar{\rho}_{(\xi)e}^2 \frac{\partial \Psi_{(\xi)}}{\partial \bar{\rho}_{(\xi)}}, \quad (5.10)$$

$$\mathbf{T}_{(\xi)} = 0. \quad (5.11)$$

The various bubble damping mechanisms [40] are included in the model by retaining the $\dot{\bar{\rho}}_{(\xi)}$ among the primitive variables.

* See DREW [39]. These terms also arise in the mixture theories of MILLS [19] and CRAINE [20].

The remaining constitutive relations are

$$(\mathbf{b}_{(\xi)}, \mathbf{q}_{(\xi)}, \dot{e}_{(\xi)}, L_{(\xi)}) = \mathbf{f}(\bar{\rho}_{(\xi)}, \dot{\bar{\rho}}_{(\xi)}, J_{(\xi)}, \theta_{(\xi)}, \mathbf{g}_{(\xi)}, \mathbf{v}_{(\xi)}, \dot{c}_{(\xi)}). \quad (5.12)$$

Equations (1.8), (1.11), (1.16), (2.5)–(2.8), (3.2), (5.7), and (5.11) are a system of equations in the unknowns λ , $\varphi_{(\xi)}$, $\bar{\rho}_{(\xi)}$, $\mathbf{v}_{(\xi)}$, $\mu_{(\xi)}$, $J_{(\xi)}$, τ , $c_{(\xi)}$, $\theta_{(\xi)}$, $\mathbf{q}_{(\xi)}$, $\mathbf{b}_{(\xi)}$, $\Psi_{(\xi)}$, $\dot{e}_{(\xi)}$, and $L_{(\xi)}$.

Since the density of the gas in the bubbles is typically small compared to the density of the liquid, the local expansional kinetic energy in this case is the kinetic energy of the liquid that is displaced as a bubble expands or contracts. This energy is conveniently written in terms of the rate of change of the density of the gas $\dot{\bar{\rho}}_{(g)}$. If the liquid is assumed to be incompressible, it is easy to show that the associated local kinetic energy coefficient can be approximated by [41]

$$I_{(g)} = \frac{R^2 \bar{\rho}_{(f)}}{3 \bar{\rho}_{(g)}^3}, \quad (5.13)$$

where R is the bubble radius and $\bar{\rho}_{(f)}$ is the density of the liquid. Furthermore, when equation (2.6) is expressed in terms of the bubble radius, it becomes

$$R \ddot{R} + \frac{3}{2}(1 - \varphi_{(g)}/\varphi_{(f)})(\dot{R})^2 = (P_{(g)} - P_{(f)})/\bar{\rho}_{(f)}. \quad (5.14)$$

When $\varphi_{(g)}/\varphi_{(f)} \rightarrow 0$, (5.14) reduces to the equation due to RAYLEIGH [42] for the growth and collapse of a bubble.

The Dusty Gas

In this formulation the dust particles are assumed to be rigid and of negligible volume. Thus we assume that

$$\Psi_{(s)} = \Psi_{(s)}(\theta_{(s)}), \quad (5.15)$$

$$\Psi_{(g)} = \Psi_{(g)}(\bar{\rho}_{(g)}, \theta_{(g)}), \quad (5.16)$$

$$\varphi_{(s)} \cong 0, \quad (5.17)$$

$$\varphi_{(g)} \cong 1, \quad (5.18)$$

where s denotes the dust and g denotes the gas.

The primitive variables are $\bar{\rho}_{(g)}$, $\mathbf{v}_{(g)}$, $\dot{\bar{\rho}}_{(g)}$, $\theta_{(g)}$, $\mathbf{g}_{(g)}$, and $\dot{c}_{(g)}$, so that, from (4.16)–(4.18),

$$P_{(s)} = 0, \quad (5.19)$$

$$P_{(g)} = \bar{\rho}_{(g)}^2 \frac{\partial \Psi_{(g)}}{\partial \bar{\rho}_{(g)}}, \quad (5.20)$$

$$\mathbf{T}_{(s)} = 0, \quad (5.21)$$

$$\mathbf{q}_{(s)} = 0. \quad (5.22)$$

The remaining constitutive relations are

$$(\mathbf{b}_{(\xi)}, \dot{e}_{(\xi)}, \mathbf{q}_{(g)}, L_{(\xi)}) = f(\bar{\rho}_{(g)}, \mathbf{v}_{(\xi)}, \dot{\mathbf{F}}_{(g)}, \theta_{(\xi)}, \mathbf{g}_{(g)}, \dot{c}_{(\xi)}). \quad (5.23)$$

Under these conditions (1.11) gives

$$\dot{\bar{\rho}}_{(g)} + \bar{\rho}_{(g)} \operatorname{div} \mathbf{v}_{(g)} = \dot{c}_{(g)}/J_{(g)}, \quad (5.24)$$

$$\dot{\rho}_{(s)} + \rho_{(s)} \operatorname{div} \mathbf{v}_{(s)} = \dot{c}_{(s)}/J_{(s)}, \quad (5.25)$$

where since $\bar{\rho}_{(s)} \gg \bar{\rho}_{(g)}$, $\rho_{(s)}$ is not negligible; equations (2.5)–(2.7) give

$$\rho_{(s)} \mathbf{a}_{(s)} = \frac{\dot{c}_{(s)}}{J_{(s)}} (\operatorname{grad} \tau - \mathbf{v}_{(s)}) + \mathbf{b}_{(s)} + \rho_{(s)} \mathbf{f}_{(s)}, \quad (5.26)$$

$$\begin{aligned} \bar{\rho}_{(g)} \mathbf{a}_{(g)} &= \frac{\dot{c}_{(g)}}{J_{(g)}} (\operatorname{grad} \tau - \mathbf{v}_{(g)}) + \rho_{(g)} \mathbf{f}_{(g)} \\ &\quad + \mathbf{b}_{(g)} - \operatorname{grad} P_{(g)} + \operatorname{div} \mathbf{T}_{(g)} \end{aligned} \quad (5.27)$$

where it has been assumed that

$$I_{(\xi)} = 0, \quad (5.28)$$

$$m_{(\xi\xi)} = 0, \quad (5.29)$$

and equation (3.2) gives

$$\rho_{(s)} \dot{\mathbf{E}}_{(s)} + E_{(s)} \frac{\dot{c}_{(s)}}{J_{(s)}} = -L_{(s)} \frac{\dot{c}_{(s)}}{J_{(s)}} + \rho_{(s)} \mathbf{r}_{(s)} + \dot{e}_{(s)}, \quad (5.30)$$

$$\begin{aligned} \bar{\rho}_{(g)} \dot{\mathbf{E}}_{(g)} + E_{(g)} \frac{\dot{c}_{(g)}}{J_{(g)}} &= \operatorname{tr}(\mathbf{T}_{(g)}^t \mathcal{L}_{(g)}) + P_{(g)} \frac{\dot{\bar{\rho}}_{(g)}}{\bar{\rho}_{(g)}} - L_{(g)} \frac{\dot{c}_{(g)}}{J_{(g)}} \\ &\quad - \operatorname{div} \mathbf{q}_{(g)} + \rho_{(g)} \mathbf{r}_{(g)} + \dot{e}_{(g)}. \end{aligned} \quad (5.31)$$

Equations (1.16), (2.8), (5.23)–(5.27), (5.30), and (5.31) are a system of equations in the unknowns $\bar{\rho}_{(g)}$, $\rho_{(s)}$, $\mathbf{v}_{(\xi)}$, $\mathbf{b}_{(\xi)}$, $\theta_{(\xi)}$, $\mathbf{q}_{(g)}$, $\dot{e}_{(\xi)}$, $c_{(\xi)}$, $L_{(\xi)}$, and τ . In the absence of chemical reactions these relations are identical to those presented by MARBLE [43].

The Fluid-Saturated Porous Solid

For a porous solid the local density $\bar{\rho}_{(s)}$ and the deformation gradient $\mathbf{F}_{(s)}$ can vary independently. Consequently, we assume

$$\Psi_{(s)} = \Psi_{(s)}(\bar{\rho}_{(s)}, \mathbf{F}_{(s)}, \theta_{(s)}), \quad (5.32)$$

$$\Psi_{(f)} = \Psi_{(f)}(\bar{\rho}_{(f)}, \theta_{(f)}), \quad (5.33)$$

where f denotes the fluid.

The primitive variables are assumed to include $\bar{\rho}_{(\xi)}$, $\mathbf{F}_{(s)}$, $\theta_{(\xi)}$, $\mathbf{g}_{(\xi)}$, $\mathbf{v}_{(\xi)}$, and $\dot{c}_{(\xi)}$, so that from (4.16) and (4.17),

$$P_{(\xi)} = \bar{\rho}_{(\xi)}^2 \frac{\partial \Psi_{(\xi)}}{\partial \bar{\rho}_{(\xi)}}, \quad (5.34)$$

$$\mathbf{T}_{(s)} = \rho_{(s)} \frac{\partial \Psi_{(s)}}{\partial \mathbf{F}_{(s)}} \mathbf{F}_{(s)}^t, \quad (5.35)$$

$$\mathbf{T}_{(f)} = \mathbf{0}. \quad (5.36)$$

The remaining constitutive relations are

$$(\mathbf{b}_{(\xi)}, \mathbf{q}_{(\xi)}, \dot{e}_{(\xi)}, L_{(\xi)}) = \mathbf{f}(\bar{\rho}_{(\xi)}, \mathbf{F}_{(s)}, \theta_{(\xi)}, \mathbf{g}_{(\xi)}, \mathbf{v}_{(\xi)}, \dot{c}_{(\xi)}). \quad (5.37)$$

Equations (1.8), (1.11), (1.16), (2.5)–(2.8), (3.2), and (5.37) are a system of equations in the unknowns $\bar{\rho}_{(\xi)}$, $\mathbf{v}_{(\xi)}$, $\mathbf{b}_{(\xi)}$, $\theta_{(\xi)}$, $\mathbf{q}_{(\xi)}$, $\dot{e}_{(\xi)}$, $c_{(\xi)}$, $L_{(\xi)}$, $\varphi_{(\xi)}$, τ , λ , and $\mu_{(\xi)} J_{(\xi)}$.

When thermal effects, chemical reactions, and local expansional kinetic energy effects are omitted, and the equations are linearized, it can be shown that these relations reduce to BIOT's equations [12]. Note from (5.33) and (5.34) that $P_{(f)}$ is the pressure in the fluid, which depends only upon $\bar{\rho}_{(f)}, \theta_{(f)}$. Similarly, from (5.32), (5.34), and (5.35), the generalized forces in the solid depend only upon the state of the solid. This is in contrast to BIOT's constitutive equations, in which the partial stresses each depend upon the states of both constituents. This resulted from his failure to include the volume fractions in his theory.

Finally, it is interesting to note that if inertial effects are neglected in the fluid, the equation of motion (2.5) becomes

$$\varphi_{(f)} \text{grad } P_{(f)} + \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} (\mathbf{v}_{(f)} - \text{grad } \tau) = \rho_{(f)} \mathbf{f}_{(f)} + \mathbf{b}_{(f)}, \quad (5.38)$$

which is a generalization of Darcy's law.

A Mixture of Ideal Gases

As a final example, the nature of the theory presented in this work can be clarified by applying it to a miscible mixture of ideal gases. In addition, this simple theory helps to illustrate the relation of the present theory to theories of miscible mixtures.

Assuming that

$$\Psi_{(\xi)} = \Psi_{(\xi)}(\rho_{(\xi)}, \theta_{(\xi)}), \quad (5.39)$$

$$I_{(\xi)} = 0, \quad (5.40)$$

$$m_{(\xi\zeta)} = 0, \quad (5.41)$$

(note that $\rho_{(\xi)}$ is used in (5.39) rather than $\bar{\rho}_{(\xi)}$) and choosing as primitive variables $\rho_{(\xi)}$, $\theta_{(\xi)}$, and $\mathbf{g}_{(\xi)}$, we obtain from (4.16)

$$P_{(\xi)} = 0. \quad (5.42)$$

Similarly from (4.8) and (4.17)

$$\mathbf{T}_{(\xi)e} = -\rho_{(\xi)e}^2 \frac{\partial \Psi_{(\xi)}}{\partial \rho_{(\xi)}} \mathbf{l}, \quad (5.43)$$

$$\mathbf{b}_{(\xi)} = 0. \quad (5.44)$$

The remaining constitutive relations are

$$(\mathbf{q}_{(\xi)}, \dot{e}_{(\xi)}) = \mathbf{f}(\rho_{(\xi)}, \theta_{(\xi)}, \mathbf{g}_{(\xi)}). \quad (5.45)$$

From (2.6), (2.7) and (5.40)–(5.42) one obtains

$$\lambda = 0. \quad (5.46)$$

Note that λ is the Lagrange multiplier associated with the volume fraction constraint (1.8).

The equation of motion (2.5) becomes

$$\rho_{(\xi)} \mathbf{a}_{(\xi)} = -\text{grad} \left(\rho_{(\xi)}^2 \frac{\partial \Psi_{(\xi)}}{\partial \rho_{(\xi)}} \right) + \rho_{(\xi)} \mathbf{f}_{(\xi)}, \quad (5.47)$$

and the energy equation (3.2) becomes

$$\rho_{(\xi)} \dot{E}_{(\xi)} = \rho_{(\xi)}^2 \frac{\partial \Psi_{(\xi)}}{\partial \rho_{(\xi)}} \frac{\dot{\rho}_{(\xi)}}{\rho_{(\xi)}} - \text{div} \mathbf{q}_{(\xi)} + \rho_{(\xi)} r_{(\xi)} + \dot{e}_{(\xi)}, \quad (5.48)$$

where it is noted that only the partial pressures $\rho_{(\xi)}^2 \frac{\partial \Psi_{(\xi)}}{\partial \rho_{(\xi)}}$ and the partial densities $\rho_{(\xi)}$ appear in the above relations. Note especially the pressure working term in (5.48). Furthermore, if $\dot{e}_{(\xi)} = 0$ and $\mathbf{q}_{(\xi)} = 0$, the equations for each constituent could be solved independently of the other constituents. This results directly from the assumed dependence on the product $\varphi_{(\xi)} \bar{\rho}_{(\xi)} = \rho_{(\xi)}$ rather than dependence on $\varphi_{(\xi)}$ and $\bar{\rho}_{(\xi)}$ separately.

Appendix A. The Virtual Mass Term

The second term in (2.1), the virtual mass term, arises as a result of local nonuniformities in the flow when the constituents of the mixture move relative to one another. It was discussed by BIOT [12] and was included by him in his theory of fluid-saturated porous media. The term is also known to be important in theories of the dynamics of fluids containing particles or bubbles [44].

Consider a fluid containing a distribution of particles. If the fluid and the particles translate with no motion relative to each other, then their kinetic energy is given by the first term in (2.1). Next, assume the fluid to be in motion with the particles held fixed. In this case, the kinetic energy of the fluid is not given correctly by the first term in (2.1) since it does not account for the nonuniformities in the flow of the fluid caused by the presence of the particles. The virtual mass term, which vanishes in the absence of relative velocity, is introduced to account for the difference. Thus, the virtual mass terms are easily

interpretable when they are introduced as kinetic energy terms in a variational formulation. Furthermore, this interpretation raises the possibility of evaluating the coefficients $m_{(\xi\zeta)}$ by experimental or analytical study of steady flow of the constituents relative to one another.

Since the flow pattern depends upon the local state of the mixture, it is assumed in this work that the coefficients $m_{(\xi\zeta)}$ depend upon the volume fractions and local densities of the constituents.

To derive the variation of the virtual mass terms, (1.22) and (1.24)–(1.26) are used to obtain

$$\begin{aligned} & \delta \int_{t_1}^{t_2} \sum_{\xi} \sum_{\zeta} \int_v \frac{1}{2} m_{(\xi\zeta)} (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) \cdot (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) dv \\ &= \int_{t_1}^{t_2} \sum_{\xi} \sum_{\zeta} \int_v [m_{(\xi\zeta)} (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) \cdot (\delta \dot{\mathbf{x}}_{(\xi)} \\ & \quad - \delta \dot{\mathbf{x}}_{(\zeta)} + \mathcal{L}_{(\zeta)} \delta \mathbf{x}_{(\zeta)}) \\ & \quad - m_{(\xi\zeta)} \mathcal{L}'_{(\xi)} (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) \cdot \delta \mathbf{x}_{(\xi)} \\ & \quad - \frac{1}{2} (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) \cdot (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) \text{grad } m_{(\xi\zeta)} \cdot \delta \mathbf{x}_{(\xi)}] dv dt, \end{aligned} \quad (\text{A.1})$$

where $\mathcal{L}_{(\xi)} = \frac{\partial \mathbf{v}_{(\xi)}}{\partial \mathbf{x}}$. By using integration by parts, we can rewrite (A.1) as

$$\begin{aligned} & \delta \int_{t_1}^{t_2} \sum_{\xi} \sum_{\zeta} \int_v \frac{1}{2} m_{(\xi\zeta)} (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) \cdot (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) dv dt \\ &= \int_{t_2}^{t_2} \sum_{\xi} \sum_{\zeta} \int_v \left\{ - (m_{(\xi\zeta)} + m_{(\zeta\xi)}) [\mathbf{a}_{(\xi)} - \mathbf{a}_{(\zeta)} \right. \\ & \quad + (\mathcal{L}'_{(\xi)} - \mathcal{L}_{(\zeta)}) (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) + (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) \text{div } \mathbf{v}_{(\xi)}] \cdot \delta \mathbf{x}_{(\xi)} \\ & \quad + \sum_{\gamma} \sum_{\zeta} \sum_{\gamma} \left[\frac{\partial m_{(\gamma\zeta)}}{\partial \bar{\rho}_{(\xi)}} \delta \bar{\rho}_{(\xi)} + \frac{\partial m_{(\gamma\zeta)}}{\partial \varphi_{(\xi)}} \delta \varphi_{(\xi)} \right. \\ & \quad \left. - \left(\frac{\partial m_{(\gamma\zeta)}}{\partial \bar{\rho}_{(\xi)}} \text{grad } \bar{\rho}_{(\xi)} + \frac{\partial m_{(\gamma\zeta)}}{\partial \varphi_{(\xi)}} \text{grad } \varphi_{(\xi)} \right) \cdot \delta \mathbf{x}_{(\xi)} \right] \\ & \quad \cdot \frac{1}{2} (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\zeta)}) \cdot (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\zeta)}) \\ & \quad \left. - \sum_{\xi} \sum_{\zeta} \overline{(m_{(\xi\zeta)} + m_{(\zeta\xi)})} |_{\mathbf{x}_{(\xi)}} (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\zeta)}) \cdot \delta \mathbf{x}_{(\xi)} \right\} dv dt. \end{aligned} \quad (\text{A.2})$$

When the $m_{(\xi\zeta)}$ are assumed to be constant, the linearized form of the right side of (A.2) introduces terms into the equations of balance of linear momentum, which are linear in the relative accelerations of the constituents. Virtual mass terms of this type appear in the mixture theories of BIOT [12], SOO [44], and others. Such terms have been criticized because they are not indifferent; that is, they do not satisfy the principle of material frame indifference.*

* See the discussion by BOWEN [10, Sec. 2.1].

However, it is easy to show that the terms

$$\mathbf{a}_{(\xi)} - \mathbf{a}_{(\xi)} + (\mathcal{L}_{(\xi)}^i - \mathcal{L}_{(\xi)})(\mathbf{v}_{(\xi)} - \mathbf{v}_{(\xi)}) \quad (\text{A.3})$$

appearing in (A.2) are indifferent, and the remainder of the terms in (A.2) are obviously indifferent. Thus, the virtual mass terms derived from the second term in (2.1) are indifferent.

That combinations such as (A.3) are indifferent was shown by ADKINS [45]. Indeed, it is pointed out by BOWEN [10] that relative acceleration terms in mixture theories should be regarded as approximations of properly indifferent forms.

A further remark is that the virtual mass forces that are introduced by (A.2) into the equations of balance of linear momentum of two given constituents are not equal and opposite, even in the case of a binary mixture (see (A.3)). In previous theories it has been assumed that the virtual mass forces are equal and opposite [46]. One consequence of the fact that they are not in the present work is that virtual mass terms appear in the definition of the stress for the mixture as a whole (equation (2.12)).

Appendix B. Variations of Specific Terms in Hamilton's Principle

The specific manipulations required to obtain the equations of motion (2.4)–(2.7) from Hamilton's principle (2.3) are outlined in this appendix. With the exception of the virtual mass term which is treated in Appendix A, the terms are treated as follows:

(1) The first kinetic energy term in (2.3) in combination with equation (1.28) gives

$$\begin{aligned} & \delta \int_{t_1}^{t_2} \sum_{\xi} \int_v \frac{1}{2} \rho_{(\xi)} \mathbf{v}_{(\xi)} \cdot \mathbf{v}_{(\xi)} dv dt \\ &= \int_{t_1}^{t_2} \sum_{\xi} \int_v \left(\rho_{(\xi)} \mathbf{v}_{(\xi)} \cdot \delta \mathbf{v}_{(\xi)} + \frac{1}{2} \mathbf{v}_{(\xi)} \cdot \mathbf{v}_{(\xi)} \frac{\delta c_{(\xi)}}{J_{(\xi)}} \right) dv dt, \end{aligned} \quad (\text{B.1})$$

which after substitution of (1.10) and (1.23) gives

$$\begin{aligned} & \delta \int_{t_1}^{t_2} \sum_{\xi} \int_v \frac{1}{2} \rho_{(\xi)} \mathbf{v}_{(\xi)} \cdot \mathbf{v}_{(\xi)} dv dt \\ &= \int_{t_1}^{t_2} \left\{ \sum_{\xi} \int_{v_0} (\hat{\rho}_{(\xi)0} + \hat{c}_{(\xi)}) \hat{\mathbf{v}}_{(\xi)} \cdot \delta \hat{\mathbf{x}}_{(\xi)} dv_0 \right. \\ & \quad \left. + \sum_{\xi} \int_v \frac{1}{2} \mathbf{v}_{(\xi)} \cdot \mathbf{v}_{(\xi)} \frac{\delta c_{(\xi)}}{J_{(\xi)}} dv \right\} dt, \end{aligned} \quad (\text{B.2})$$

where a transformation of the integral to the reference configuration has been used. Then after integration by parts one obtains

$$\begin{aligned}
& \delta \int_{t_1}^{t_2} \sum_{\xi} \int_v \frac{1}{2} \rho_{(\xi)} \mathbf{v}_{(\xi)} \cdot \mathbf{v}_{(\xi)} dv dt \\
&= \int_{t_1}^{t_2} \sum_{\xi} \int_v \left[\left(-\rho_{(\xi)} \mathbf{a}_{(\xi)} - \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} \mathbf{v}_{(\xi)} \right) \cdot \delta \mathbf{x}_{(\xi)} \right. \\
&\quad \left. + \frac{1}{2} \mathbf{v}_{(\xi)} \cdot \mathbf{v}_{(\xi)} \frac{\delta c_{(\xi)}}{J_{(\xi)}} \right] dv dt; \tag{B.3}
\end{aligned}$$

(2) The third kinetic energy term gives

$$\begin{aligned}
& \delta \int_{t_1}^{t_2} \sum_{\xi} \int_v \frac{1}{2} \rho_{(\xi)} I_{(\xi)} \dot{\bar{\rho}}_{(\xi)}^2 dv dt \\
&= \int_{t_1}^{t_2} \sum_{\xi} \left\{ \int_{v_0} (\hat{\rho}_{(\xi)0} + \hat{c}_{(\xi)}) \delta \left(\frac{1}{2} \hat{I}_{(\xi)} \dot{\bar{\rho}}_{(\xi)}^2 \right) dv_0 \right. \\
&\quad \left. + \int_v \frac{1}{2} I_{(\xi)} \dot{\bar{\rho}}_{(\xi)}^2 \frac{\delta c_{(\xi)}}{J_{(\xi)}} dv \right\} dt \\
&= \int_{t_1}^{t_2} \sum_{\xi} \int_v \left\{ \left[-\rho_{(\xi)} \overline{\left(\frac{\partial I_{(\xi)}}{\partial \bar{\rho}_{(\xi)}} \right)} - \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} I_{(\xi)} \dot{\bar{\rho}}_{(\xi)} \right. \right. \\
&\quad \left. \left. + \sum_{\xi} \frac{1}{2} \rho_{(\xi)} \frac{\partial I_{(\xi)}}{\partial \bar{\rho}_{(\xi)}} \dot{\bar{\rho}}_{(\xi)}^2 \right] \delta \bar{\rho}_{(\xi)} \right. \\
&\quad \left. + \sum_{\xi} \frac{1}{2} \rho_{(\xi)} \frac{\partial I_{(\xi)}}{\partial \varphi_{(\xi)}} \dot{\bar{\rho}}_{(\xi)}^2 \delta \varphi_{(\xi)} \right. \\
&\quad \left. + \frac{1}{2} I_{(\xi)} \dot{\bar{\rho}}_{(\xi)}^2 \frac{\delta c_{(\xi)}}{J_{(\xi)}} + \left[\frac{1}{2} \rho_{(\xi)} \dot{\bar{\rho}}_{(\xi)}^2 \text{grad } I_{(\xi)} \right. \right. \\
&\quad \left. \left. - \sum_{\xi} \frac{1}{2} \rho_{(\xi)} \dot{\bar{\rho}}_{(\xi)}^2 \left(\frac{\partial I_{(\xi)}}{\partial \bar{\rho}_{(\xi)}} \text{grad } \bar{\rho}_{(\xi)} + \frac{\partial I_{(\xi)}}{\partial \varphi_{(\xi)}} \text{grad } \varphi_{(\xi)} \right) \right] \cdot \delta \mathbf{x}_{(\xi)} \right\} dv dt; \tag{B.4}
\end{aligned}$$

(3) The first virtual work term becomes

$$\int_{t_1}^{t_2} \sum_{\xi} \int_v -\text{tr}(\mathbf{T}_{(\xi)}' \text{grad } \delta \mathbf{x}_{(\xi)}) dv dt = \int_{t_1}^{t_2} \sum_{\xi} \int_v \text{div } \mathbf{T}_{(\xi)} \cdot \delta \mathbf{x}_{(\xi)} dv dt \tag{B.5}$$

by using integration by parts;

(4) The first constraint condition gives

$$\delta \left(\sum_{\xi} \varphi_{(\xi)} - 1 \right) |_{\mathbf{x}} = \sum_{\xi} (\delta \varphi_{(\xi)} - \text{grad } \varphi_{(\xi)} \cdot \delta \mathbf{x}_{(\xi)}) \tag{B.6}$$

where (1.25) has been used;

(5) The second constraint condition gives

$$\begin{aligned} & \int_{t_1}^{t_2} \int_v \tau \delta \left(\sum_{\xi} \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} \right) \Big|_x dv dt \\ &= \int_{t_1}^{t_2} \int_v \sum_{\xi} \tau \left[\frac{\delta \dot{c}_{(\xi)}}{J_{(\xi)}} - \text{grad} \left(\frac{\dot{c}_{(\xi)}}{J_{(\xi)}} \right) \cdot \delta \mathbf{x} - \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} \text{div} \delta \mathbf{x}_{(\xi)} \right] dv dt \\ &= \int_{t_1}^{t_2} \int_v \sum_{\xi} \left(-\dot{\tau} \Big|_{\mathbf{x}_{(\xi)}} \frac{\delta c_{(\xi)}}{J_{(\xi)}} + \frac{\dot{c}_{(\xi)}}{J_{(\xi)}} \text{grad} \tau \cdot \delta \mathbf{x}_{(\xi)} \right) dv dt; \end{aligned}$$

(6) The last constraint condition gives

$$\begin{aligned} & \int_{t_1}^{t_2} \sum_{\xi} \int_v \mu_{(\xi)} \delta \left(J_{(\xi)} - \frac{\rho_{(\xi)0} + c_{(\xi)}}{\rho_{(\xi)}} \right) dv dt \\ &= \int_{t_1}^{t_2} \sum_{\xi} \int_v \left[-\text{grad} \mu_{(\xi)} J_{(\xi)} \cdot \delta \mathbf{x}_{(\xi)} \right. \\ & \quad \left. - \frac{\mu_{(\xi)} J_{(\xi)}}{\rho_{(\xi)}} \frac{\delta c_{(\xi)}}{J_{(\xi)}} + \mu_{(\xi)} J_{(\xi)} \left(\frac{\delta \bar{\rho}_{(\xi)}}{\bar{\rho}_{(\xi)}} + \frac{\delta \varphi_{(\xi)}}{\varphi_{(\xi)}} \right) \right] dv dt, \quad (\text{B.8}) \end{aligned}$$

where equation (1.24) has been used.

Appendix C. Constraint Conditions for the Entropy Inequality

The volume fraction constraint (1.8), the mass transfer constraint (1.16), and the constituent mass conservation equations (1.10) place restrictions upon the terms in the entropy inequality (4.5) that can be varied independently. For example, even if the $\dot{c}_{(\xi)}$ were not included in the list of primitive variables, it is clear that (1.10) would still not allow one to argue that all of the $\dot{c}_{(\xi)}$ in (4.5) could be varied independently in the manner required by the procedure described by COLEMAN & NOLL [36]. In a similar fashion the time derivatives and the gradients of (1.8), (1.10), and (1.16) place restrictions upon the constitutive variables that can be varied independently.

In this appendix these constraint conditions will be investigated. The constraint equations and the appropriate derivatives of these relations will first be listed. The volume fraction constraint is

$$\sum_{\xi} \varphi_{(\xi)} = 1. \quad (\text{C.1})$$

The gradient of (C.1) is

$$\sum_{\xi} \mathbf{H}_{(\xi)} = \mathbf{0}, \quad (\text{C.2})$$

where $\mathbf{H}_{(\xi)} = \text{grad} \varphi_{(\xi)}$. The material derivative of (C.1) following the motion of the γ^{th} constituent is

$$\sum_{\xi} [\dot{\varphi}_{(\xi)} + \mathbf{H}_{(\xi)} \cdot (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\xi)})] = 0, \quad (\text{C.3})$$

where (1.6) has been used. Similarly the material derivative of (C.2) is

$$\sum_{\xi} \left[\dot{\mathbf{H}}_{(\xi)} + \frac{\partial \mathbf{H}_{(\xi)}}{\partial \mathbf{x}} (\mathbf{v}_{(\delta)} - \mathbf{v}_{(\xi)}) \right] = \mathbf{0}. \quad (\text{C.4})$$

Finally, the material derivative of (C.3) is

$$\begin{aligned} \sum_{\xi} \left\{ \dot{\phi}_{(\xi)} + (\dot{\mathbf{H}}_{(\xi)} + \mathcal{L}'_{(\xi)} \mathbf{H}_{(\xi)}) \cdot (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\gamma)}) \right. \\ \left. + \left[\dot{\mathbf{H}}_{(\xi)} + \frac{\partial \mathbf{H}_{(\xi)}}{\partial \mathbf{x}_{(\xi)}} (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\xi)}) \right] \cdot (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\xi)}) \right. \\ \left. + \mathbf{H}_{(\xi)} \cdot [\dot{\mathbf{v}}_{(\gamma)} - \dot{\mathbf{v}}_{(\xi)} + (\mathcal{L}_{(\gamma)} - \mathcal{L}_{(\xi)}) (\mathbf{v}_{(\xi)} - \mathbf{v}_{(\xi)})] \right\} = 0. \end{aligned} \quad (\text{C.5})$$

Next the mass transfer constraint is

$$\sum_{\xi} \dot{c}_{(\xi)} / J_{(\xi)} = 0. \quad (\text{C.6})$$

The gradient of this relation is

$$\sum_{\xi} [\dot{\mathbf{K}}_{(\xi)} / J_{(\xi)} + \mathcal{L}'_{(\xi)} \mathbf{K}_{(\xi)} / J_{(\xi)} - (\dot{c}_{(\xi)} / J_{(\xi)}^2) \text{grad } J_{(\xi)}] = \mathbf{0}, \quad (\text{C.7})$$

where $\mathbf{K}_{(\xi)} = \text{grad } c_{(\xi)}$. The material derivative of (C.7) following the γ^{th} constituent is

$$\begin{aligned} \sum_{\xi} \{ [\dot{c}_{(\xi)} + \dot{\mathbf{K}}_{(\xi)} \cdot (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\xi)}) + \mathcal{L}'_{(\xi)} \mathbf{K}_{(\xi)} \cdot (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\xi)})] / J_{(\xi)} \\ - (\dot{c}_{(\xi)} / J_{(\xi)}^2) [J_{(\xi)} + \text{grad } J_{(\xi)} \cdot (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\xi)})] \} = 0. \end{aligned} \quad (\text{C.8})$$

The last set of constraints comes from the mass conservation relations:

$$\bar{\rho}_{(\xi)} \varphi_{(\xi)} J_{(\xi)} = \bar{\rho}_{(\xi)0} \varphi_{(\xi)0} + c_{(\xi)}. \quad (\text{C.9})$$

The gradient of (C.9) is

$$\mathbf{G}_{(\xi)} \varphi_{(\xi)} J_{(\xi)} + \bar{\rho}_{(\xi)} \mathbf{H}_{(\xi)} J_{(\xi)} + \bar{\rho}_{(\xi)} \varphi_{(\xi)} \text{grad } J_{(\xi)} = \mathbf{K}_{(\xi)}, \quad (\text{C.10})$$

where $\mathbf{G}_{(\xi)} = \text{grad } \bar{\rho}_{(\xi)}$. The material time derivative of (C.9) is

$$\dot{\bar{\rho}}_{(\xi)} \varphi_{(\xi)} J_{(\xi)} + \bar{\rho}_{(\xi)} \dot{\varphi}_{(\xi)} J_{(\xi)} + \bar{\rho}_{(\xi)} \varphi_{(\xi)} \dot{J}_{(\xi)} = \dot{c}_{(\xi)}. \quad (\text{C.11})$$

The second time derivative of (C.9) is

$$\begin{aligned} \ddot{\bar{\rho}}_{(\xi)} \varphi_{(\xi)} J_{(\xi)} + 2 \dot{\bar{\rho}}_{(\xi)} \dot{\varphi}_{(\xi)} J_{(\xi)} + 2 \bar{\rho}_{(\xi)} \dot{\varphi}_{(\xi)} \dot{J}_{(\xi)} \\ + 2 \bar{\rho}_{(\xi)} \dot{\varphi}_{(\xi)} \dot{J}_{(\xi)} + \bar{\rho}_{(\xi)} \ddot{\varphi}_{(\xi)} J_{(\xi)} + \rho_{(\xi)} \varphi_{(\xi)} \ddot{J}_{(\xi)} = \ddot{c}_{(\xi)}. \end{aligned} \quad (\text{C.12})$$

Finally the time derivative of (C.10) is

$$\begin{aligned} & \dot{\mathbf{G}}_{(\xi)} \varphi_{(\xi)} \mathbf{J}_{(\xi)} + \mathbf{G}_{(\xi)} \dot{\varphi}_{(\xi)} \mathbf{J}_{(\xi)} + \mathbf{G}_{(\xi)} \varphi_{(\xi)} \dot{\mathbf{J}}_{(\xi)} \\ & + \dot{\bar{\rho}}_{(\xi)} \mathbf{H}_{(\xi)} \mathbf{J}_{(\xi)} + \bar{\rho}_{(\xi)} \dot{\mathbf{H}}_{(\xi)} \mathbf{J}_{(\xi)} + \bar{\rho}_{(\xi)} \mathbf{H}_{(\xi)} \dot{\mathbf{J}}_{(\xi)} \\ & + \dot{\bar{\rho}}_{(\xi)} \varphi_{(\xi)} \text{grad } \mathbf{J}_{(\xi)} + \bar{\rho}_{(\xi)} \dot{\varphi}_{(\xi)} \text{grad } \mathbf{J}_{(\xi)} \\ & + \bar{\rho}_{(\xi)} \varphi_{(\xi)} \overline{\dot{\text{grad } \mathbf{J}_{(\xi)}}} = \dot{\mathbf{K}}_{(\xi)}. \end{aligned} \quad (\text{C.13})$$

Equations (C.1) through (C.13) are the constraint conditions which must be considered when applying COLEMAN & NOLL's arguments to (4.5). To do this the constraint conditions are introduced into (4.5) by the method of Lagrange multipliers [47]. The multipliers for these relations are denoted by D^1 , D^2 , D^3 , D^4 , $D^5_{\zeta\gamma}$, D^6 , D^7 , D^8_γ , D^9 , D^{10} , D^{11} , D^{12} , and D^{13} , respectively.

Some of these multiplier terms do not place constraints upon the terms in (4.5). For example, only (C.4) and (C.5) have terms which depend on $\partial \mathbf{H}_{(\xi)} / \partial \mathbf{x}$. Therefore, one concludes that

$$\mathbf{D}^4 \cdot (\mathbf{v}_{(\delta)} - \mathbf{v}_{(\xi)}) + D^5_{\zeta\gamma} (\mathbf{v}_{(\zeta)} - \mathbf{v}_{(\xi)}) \cdot (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\xi)}) = 0. \quad (\text{C.14})$$

For the case when $\zeta = \xi \neq \delta$, one obtains

$$\mathbf{D}^4 = \mathbf{0}, \quad (\text{C.15})$$

and therefore

$$D^5_{\zeta\gamma} = 0. \quad (\text{C.16})$$

Thus, (C.4) and (C.5) do not give rise to any additional constraints. Having eliminated (C.4) and (C.5), we notice that $\dot{\varphi}_{(\xi)}$ appears only in (C.12). Since none of the terms in (4.5) depend on $\dot{\varphi}_{(\xi)}$, it is concluded that

$$D^{12} = 0. \quad (\text{C.17})$$

A similar argument based on $\overline{\dot{\text{grad } \mathbf{J}_{(\xi)}}}$ in (C.13) gives

$$D^{13} = 0. \quad (\text{C.18})$$

Among the remaining constraint equations, $\dot{\mathbf{K}}_{(\xi)}$ appears only in (C.7) and (C.8), giving

$$\mathbf{D}^7 + D^8_\gamma (\mathbf{v}_{(\gamma)} - \mathbf{v}_{(\xi)}) = \mathbf{0}. \quad (\text{C.19})$$

By considering the case $\gamma = \xi$, we can then conclude that

$$\mathbf{D}^7 = \mathbf{0}; \quad D^8_\gamma = 0. \quad (\text{C.20})$$

The remaining constraint conditions do not depend on $\dot{\theta}_{(\xi)}$, $\dot{\bar{\rho}}_{(\xi)}$, $\dot{\mathbf{F}}_{(\xi)}$, $\dot{\mathbf{c}}_{(\xi)}$, or $\dot{\mathbf{G}}_{(\xi)}$, and thus the coefficients of these terms in (4.5) may be equated to zero.

Next, since $\mathbf{K}_{(\xi)}$ now appears only in (C.10),

$$\mathbf{D}^{10} = \mathbf{0}. \quad (\text{C.21})$$

Then because $H_{(\xi)}$ appears only (C.2) and (C.3)

$$D^2 + D_\gamma^3(v_{(\gamma)} + v_{(\xi)}) = 0 \quad (\text{C.22})$$

and, therefore,

$$D^2 = 0; \quad D_\gamma^3 = 0. \quad (\text{C.23})$$

Now $\phi_{(\xi)}$ appears only in (C.11) and, thus,

$$D^{11} = 0. \quad (\text{C.24})$$

The remaining constraint relations are (C.1), (C.6), and (C.9). Thus, one sees that (4.5) is affected only by the primary constraint conditions; the gradients and material derivatives do not give rise to additional conditions. Furthermore, in the list of primitive variables—with the exception of $\dot{c}_{(\xi)}$ —the material time derivatives of the primitive variables are not constrained.

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Note Added in Proof. In this work we have used a single entropy inequality for the mixture and have introduced an immiscibility postulate which states that the free energy of a given constituent of an immiscible mixture is a function only of the constitutive variables associated with that constituent. SAMPAIO & WILLIAMS [SAMPAIO, R., & W. O. WILLIAMS, Thermodynamics of diffusing mixtures. *J. de Mécanique* **18**, 19–45 (1979)] postulate an entropy inequality for each constituent of an immiscible mixture. For two special non-reacting examples (a mixture of two linearly viscous incompressible fluids and a mixture of a linearly viscous incompressible fluid and an elastic solid), they then prove that the immiscibility postulate follows as a consequence of their entropy inequalities.

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